

Training-Free Depth Pruning and Post-Hoc Error Compensation for 1-Bit Large Language Models

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Abstract

Extreme weight quantization has produced openly available models that retain 27B-class capability at 1.125 bits per weight. We study whether such models remain structurally compressible after quantization, under three constraints: no training or fine-tuning, no calibration-based updates to surviving weights, and a total compute budget of a single 64 GB consumer laptop. On a public 1-bit 27B hybrid-attention model, we make five observations. (1) Depth redundancy persists after extreme quantization: removing four transformer blocks and three attention sublayers reduces the language model by 15.4% (709 MB) at a cost of 4.5 points of macro accuracy over a 500-item, four-task evaluation suite, with GSM8K unchanged at reference level; pruning-induced degradation composes near-additively across removed components. (2) The quantization format itself is redundant: stored zero-points satisfy $z = \text{fl6}(-s/2)$ for every quantization group, so they can be recomputed inside the inference kernels, reducing resident memory by a measured 420 MB with numerically identical logits. (3) Second-order quantization of the scale parameters to 8 bits is loss-free within evaluation noise, yielding a combined 3.82 GB model (-18.9%); at 4 bits, accuracy degrades sharply. (4) A closed-form error-compensation procedure, which absorbs a linear approximation of the removed layers’ contribution into three surviving projections and requantizes them into the native 1-bit format, recovers +1.0 macro points at identical model size; we characterize the boundaries of the regime in which this holds. (5) Across six controlled cases, calibration-based proxy metrics (full-vocabulary teacher-forced KL divergence, measured in two data regimes) fail to predict end-task outcomes, including a search procedure that dominates the proxy frontier yet underperforms a hand-designed baseline on the evaluation suite. We conclude that proxy quality metrics are reliable only within a single operator family, and that end-task evaluation must remain the acceptance criterion for post-training compression.

1 Introduction

Inference on memory-constrained devices is dominated by weight movement: at the small batch sizes typical of interactive use, decoding throughput is bounded by the bytes streamed per generated token [2]. This constraint has motivated progressively more aggressive weight quantization, culminating in binary and ternary formats [5, 6] and, recently, in openly released models that preserve most of their full-precision benchmark performance at 1.125 bits per weight. Such models are commonly regarded as terminal artifacts: the quantization procedure is assumed to have consumed the available redundancy, and further compression is expected to require retraining.

In this work we examine what redundancy remains *after* extreme quantization. We adopt deliberately restrictive rules: no training or fine-tuning of any kind, no calibration-based updates to surviving weights, and, for the primary results, surviving weights that are byte-for-byte identical to the released checkpoint. These rules serve three purposes. First, they make every measured accuracy change attributable solely to the structural intervention. Second, they make claims directly verifiable through byte comparison, logit equality, and per-item reproduction of evaluation runs. Third, they reflect a practical property of the 1-bit setting: the released weights are the product of a proprietary conversion procedure that downstream users cannot reproduce, so modifications to weight values carry unrecoverable risk. All experiments were conducted on a single 64 GB Apple Silicon laptop.

Our contributions are summarized in Table 1.

Contribution	Result
Depth-redundancy analysis at four granularities	near-additive composition of degradation; superadditivity factor 1.09–1.13 $\times$; replicated on a second architecture (§4, §9)
Pruned model, unmodified surviving weights	−709 MB (−15.4%); macro .7963 vs. .8413; GSM8K unchanged (§4)
Zero-point reparameterization	$z = \text{f16}(-s/2)$ verified for all groups; −420,225,024 B resident memory, logits unchanged (§5)
Post-pruning error compensation	closed-form update; 1% of sign bits changed; +1.0 macro at identical size (§7)
Proxy-metric reliability study	six controlled failures of KL-based selection; per-operator-family degradation slopes (§8)

Table 1: Summary of contributions.

2 Related Work

Post-training quantization. GPTQ [1] and AWQ [2] quantize pretrained models to 3–4 bits using second-order information and activation-aware scaling, respectively; SparseGPT [3] extends closed-form error compensation to pruning. Our compensation procedure (§7) is closest in spirit to this line, but operates after *layer removal* rather than per-weight rounding, and projects its update back into a binary format. Double quantization of scale parameters appears in QLoRA [4]; we measure its accuracy knee on a 1-bit base model. Binary and ternary pretraining is developed in the BitNet line [5, 6]; the model we study instead derives from a pretrained full-precision network.

Structural pruning of LLMs. Depth pruning by layer-importance ranking is studied by ShortGPT [7], which introduced the Block Influence metric we use for shortlisting, and by Gromov et al. [8], who report that deep layers of full-precision models are largely removable. LaCo [9] merges adjacent layers; Sheared-LLaMA [10] and LLM-Pruner [11] combine structured pruning with retraining. Attention-head redundancy in full-precision transformers was established by Michel et al. [12]. We evaluate the training-free portions of this toolbox on an extremely quantized model and find that several of its standard conclusions (deep-layer removability, merging, head pruning) do not transfer (§4–§8).

Proxy metrics for compression. Jaiswal et al. [13] observe that perplexity understates the capability loss of compressed LLMs on harder tasks. Our results extend this concern to full-vocabulary teacher-forced KL under two data regimes, and localize it: proxy reliability depends on the operator family rather than on the metric alone.

3 Experimental Setup

Model. We study a publicly released 27B-parameter hybrid-attention causal language model quantized to a binary format with group-wise affine parameters: each weight takes the value $\pm s_g/2$, with one f16 scale s_g per group of 128 weights, giving 1.125 bits per weight and 4.71 GB of language-model parameters. The network has 64 blocks (75% linear attention, 25% softmax attention with GQA), hidden size 5120, and a 248K vocabulary. A dense sibling model (8B, 36 blocks, softmax attention only, same format) is used to test generalization across architectures.

Evaluation protocol. We employ a three-level protocol (Table 2). Proxy screening computes full-vocabulary teacher-forced forward KL divergence against the unmodified model in two regimes: *on-policy*, over the model’s own sampled reasoning traces, and *off-policy*, over held-out natural text. Candidates that survive screening are confirmed at 100 probes per regime. Final decisions rest exclusively on a 500-item evaluation suite spanning GSM8K, MATH-500, IFEval, and MMLU-Redux, generated with the model’s recommended sampling configuration in reasoning mode and scored with rule-based extraction; items are individually seeded for reproducibility. Calibration data (64 sequences) is disjoint from all probe sets, and no procedure is ever fitted on probe data.

Compression operators. Structural operators remove whole transformer blocks, individual attention or MLP sublayers, or whole GQA key–value groups (six query heads, one KV head, and the associated

Level	Metric	Size	Role
Screening	teacher-forced KL, 2 regimes	24 probes/regime	candidate shortlisting
Confirmation	teacher-forced KL, 2 regimes	100 probes/regime	shortlist verification
End-task	4-task accuracy	500 items	acceptance criterion

Table 2: Evaluation protocol. Both KL regimes are required; §8 documents failure cases of single-regime and cross-family comparisons.

group-aligned output-projection columns); all surviving tensors are unmodified. Two weight-modifying procedures (§6, §7) are reported separately from the unmodified-weight results throughout.

4 Depth Redundancy and Layer Pruning

We first score every block by individual removal. The resulting profile is consistent across granularities: the first and final blocks are essential (removing either causes severe degradation); redundancy concentrates at intermediate depth; and the two KL regimes disagree on the ranking of intermediate blocks (one block appears removable on-policy and highly damaging off-policy; case 1 of §8). The same profile recurs at sublayer granularity, where deep linear-attention sublayers are individually inexpensive to remove and the attention and MLP components of a block can be removed independently, and at head granularity, where early softmax-attention blocks contain KV groups whose removal is nearly free under the proxy metric.

Degradation composes near-additively: over operator types and both models, the KL of a removal set is $0.97\text{--}1.20\times$ the sum of its constituents, with the superadditivity factor increasing for same-type combinations. Table 3 reports the end-task results. The composed configuration (four blocks and three deep attention sublayers) achieves twice the reduction of the six-block baseline at two points less degradation. Degradation is strongly task-dependent: GSM8K accuracy is unaffected by the final configuration, while MMLU-Redux accounts for the largest share of the loss, indicating that depth pruning primarily removes diffuse knowledge capacity rather than multi-step reasoning ability.

Configuration	−MB	GSM8K	MATH500	IFEval	MMLU-R	Macro
unmodified baseline	0	.910	.720	.930	.805	.8413
2 blocks removed	118	.920	.660	.880	.790	.8125
4 blocks removed	236	.910	.680	.920	.760	.8175
6 blocks removed	360	.880	.640	.860	.725	.7762
8 blocks removed	478	.870	.720	.790	.620	.7500
4 blocks + 3 attn. sublayers	709	.910	.670	.880	.725	.7963
evolutionary-search selection (§8)	365	.880	.610	.830	.730	.7625
+ 8 KV groups (§8)	739	.850	.630	.830	.735	.7612

Table 3: End-task results for the unmodified-weight configurations (500 items; sizes relative to the 4.71 GB checkpoint). Rows below the rule were selected by proxy metrics and rejected by end-task evaluation. The final pruned checkpoint reproduces its evaluated configuration on all 500 items.

Several further axes permitted under our constraints were evaluated and rejected (Table 4).

5 Zero-Point Reparameterization

The released checkpoint stores, per quantization group, both a scale and a zero-point. The binary mapping constrains the zero-point to $z = \text{fl6}(-s/2)$, an identity we verified over every group in the model; the model card acknowledges the redundancy and defers its removal to future runtime support. Deleting the stored zero-points reduces the checkpoint by 420 MB with unchanged logits. We further implement the deferred runtime support: a modified kernel path recomputes the zero-point in-register during the fused dequantize-multiply. Outputs match the stored-zero-point path on all matmul and dequantization routes, on

Axis	Evaluation	Outcome
width pruning	per-channel magnitude statistics	no prunable low-magnitude population at 1 bit
vocabulary (input)	held-out OOV mass	heavy tail; rows are required
vocabulary (output)	emission-mass row selection	teacher-forced logits unchanged, generation degraded: +26% length, $2.3\times$ truncation rate (case 2, §8)
layer merging	element-wise sign vote; 2-bit union merge	both inferior to plain removal at matched size
sign bits	entropy estimate	8.000 bits/byte; incompressible

Table 4: Negative results under the unmodified-weight constraint.

synthetic inputs and on the full model, and resident memory decreases by the full 420,225,024 bytes. Because the recomputation replaces a memory stream with a single multiply in bandwidth-bound kernels, throughput is not reduced.

6 Second-Order Quantization of Scale Parameters

After reparameterization, the model consists of 3.77 GB of sign bits and 420 MB of f16 scales. Following the double-quantization approach of Dettmers et al. [4], we quantize the scales with per-row affine parameters (row groups of 32–40 entries). This modifies effective weight values and is therefore reported separately from the unmodified-weight results (Table 5). At 8 bits, degradation is within evaluation noise and composes cleanly with depth pruning, yielding a 3.82 GB model (-18.9%) at .7900 macro. At 4 bits, accuracy drops by 2.9 points even though the proxy metric remains $76\times$ below that of the mildest block removal (case 4 of §8).

Configuration	–MB	Macro	Assessment
8-bit scales	612	.8363	within noise (-0.5)
pruned + 8-bit scales	888	.7900	3.82 GB; additive with pruning
4-bit scales	717	.8125	accuracy knee (-2.9)
pruned + compensation (§7)	709	.8063	+1.0 at identical size
pruned + 8-bit scales + compensation	888	.7825	negative interaction (case 5)
6-block pruning + compensation	360	.7750	no effect; update norm saturates

Table 5: End-task results for weight-modifying procedures (500 items). Sizes relative to the 4.71 GB checkpoint.

7 Post-Pruning Error Compensation

Layer removal deletes the removed modules’ contributions to the residual stream. We compensate for this loss without training and without leaving the native format. For each removed group of layers, we estimate a linear approximation of its contribution: a ridge regression from the input of a designated surviving downstream projection to the residual-stream deficit, with normal equations accumulated over the 64-sequence calibration set. The resulting update ΔW is constrained to $\|\Delta W\| \leq 0.25 \|W\|$ so that requantization error remains second-order, added to the dequantized weights of the designated projection, and requantized into the binary format (per group: scale $2^{\lceil w^* \rceil}$, signs $\text{sign}(w^*)$). The procedure changes no tensor other than the designated projections and adds no parameters.

Applied to the pruned model, the update changes 0.8–1.7% of sign bits across three projections, perturbs scales by under 4.5%, and improves macro accuracy by +1.0 points at identical model size (Table 5), with gains distributed across tasks and no degenerate generation behavior. To the best of our knowledge, this is the first demonstration that a pruned 1-bit model can be improved by a closed-form update expressed

entirely within its own quantization format. The applicable regime is narrow, and we report its boundaries. A diagonal (per-channel gain) variant fails, which we attribute to downstream RMSNorm layers already absorbing static scale effects. Applying the procedure on top of scale quantization reverses its sign despite a confirmation-level proxy profile indistinguishable from the successful run (case 5). On a deeper six-block pruning, the norm constraint saturates at all fitting sites and the measured effect is null.

8 Proxy Metrics versus End-Task Evaluation

Table 6 lists six controlled cases, under one fixed protocol, in which proxy metrics mispredicted end-task outcomes.

#	Configuration	Proxy assessment	End-task outcome
1	single intermediate block	inexpensive (on-policy only)	severe off-policy degradation
2	vocabulary row selection	logits unchanged	+26% length, $2.3\times$ truncations
3	evolutionary-search selection	dominates frontier, both regimes	−1.4 vs. size-matched baseline
4	4-bit scales	$76\times$ below mildest removal	−2.9; matches 2-block removal
5	compensation after scale quant.	matches successful run’s profile	sign of effect reversed
6	8 KV groups on pruned model	subadditive; below 6-block config	−3.5 for 30 MB

Table 6: Failures of proxy-based selection. Case 3: a $(1+\lambda)$ evolutionary search over 311 per-block configurations whose KL advantage over hand-designed baselines grows to 40% off-policy at the largest size tier, and whose selected configuration nonetheless underperforms on the evaluation suite; the search overfit its proxy objective.

Cases 3–6 support a consistent interpretation: operator families exhibit characteristic ratios between proxy divergence and end-task degradation. Removal of redundant structure has a small ratio; the removed computation is substitutable, and KL ranks such configurations in agreement with end-task results. Operators that perturb every token’s computation by a small amount (scale quantization noise, KV-group removal, requantization residue) have a large ratio: teacher-forced divergence evaluates isolated next-token predictions and does not capture the compounding of dense perturbations over long autoregressive generations. In line with the concerns of Jaiswal et al. [13], we conclude that proxy comparisons are meaningful only within an operator family, and that no proxy, regardless of prior validation, should replace end-task evaluation as the acceptance criterion.

9 Cross-Architecture Generalization

Table 7 repeats the core analyses on the dense 8B sibling. The qualitative structure transfers: boundary layers are essential, redundancy concentrates at intermediate depth, the two proxy regimes disagree, and composition is near-additive in the same range. The magnitude of removable redundancy does not transfer: the least damaging single removal costs $6\times$ more, and the four-block recipe that costs 2.4 points on the 27B costs 14.8 points on the 8B. An interpolation test rejects a shared mapping from proxy divergence to degradation across models. We conclude that depth-prunability reflects over-provisioned depth relative to capability, while the format-level redundancy of §5 (verified on all 254 quantized tensors of the sibling) is a property of the model family.

10 Limitations

Our headline results concern one model family and one quantization format. With $N=500$ per evaluation run, resolution is approximately ± 1.8 macro points at one standard deviation, so effects near one point are directional rather than definitive on single-run evidence. The reasoning-mode token budget interacts with the degradation floor. The six proxy failures are controlled demonstrations under one protocol rather than a survey. The compensation result is an existence proof with characterized boundaries, not a general

Property	27B (hybrid)	8B (dense)
boundary layers essential	yes	yes
intermediate-depth redundancy	yes	yes
regime disagreement	yes	yes
composition factor	1.09–1.13×	0.97–1.20×
min. single-removal KL (on-policy)	.0033	.0198
4-block removal, end-task cost	−2.4	−14.8
shared proxy-to-degradation mapping	rejected (predicted −7 to −9; measured −4.3)	

Table 7: Generalization to a second architecture.

method. Finally, the single-machine budget bounded search breadth, sampling repeats, and evaluation sizes throughout.

11 Conclusion

A model that had already undergone $14\times$ weight compression retained: a fifth of its bytes removable or reparameterizable at a cost of at most 4.5 macro points with mathematical reasoning intact; a 420 MB plane of derivable quantization parameters; loss-free second-order quantization of its scales at 8 bits and a sharp knee below; and a neighborhood in its own sign/scale space containing strictly better models, reachable by one closed-form solve. None of this required training. It did require treating every proxy metric as a hypothesis generator and end-task evaluation as the sole arbiter. Code and artifacts will be released upon publication.

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